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Research Paper

A Great and Engaging Title about RXJ1720+2638

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Abstract

- Radio minihalos are pretty neat things.
- Present new 8–12 and archival 1–2 GHz VLA images of RXJ1720+2638.
- We find that the spectral power follows a single power-law to high-frequency. Splitting into central and tail regions, we find that the tail is brighter than expected.
- Our results are in line with previous papers, offer the first observational evidence at high frequencies.
- Empirical evidence towards spectral steepening along the tail, though more effort required here.
- The single power-law favours the hadronic scenario.

Keywords: Key1, Key2, Key3, Key4

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Galaxy clusters are the largest gravitationally bound objects we have observed within the universe, and are host to a rich variety of astronomical phenomena. With total masses typically of order $M_{500} \sim 10^{14}$–10^{15} $M_{\odot}$, a dominant $\sim 85\%$ of a clusters mass is attributed to dark matter. Approximately 3% comes from the baryonic matter of the galaxies themselves. The remaining $\sim 12\%$ of the mass budget comes from the intracluster medium (ICM), a hot diffuse plasma ($T \sim 10^7$–10^8 K, $n_e \sim 10^3$ m^{-3}) spread throughout the cluster. Clusters typically grow from the accretion of matter from the cosmic web, or via mergers with other clusters (e.g. Kravtsov & Borgani 2012).

Merger shocks and turbulence introduce massive amounts of energy into the ICM, causing a acceleration and heating of the particles and, subsequently, cooling via X-ray bremsstrahlung radiation. Particles may also be accelerated to the relativistic level, becoming cosmic ray electrons (CRe). These move within the large scale magnetic fields induced by the ionisation of the ICM, generating synchrotron emission at radio frequencies (e.g Feretti et al. 2012). Interactions with low energy ICM photons are also expected to produce inverse Compton radiation as hard X-rays (e.g. Govoni & Feretti 2004). However, these are more difficult to detect due to the dominance of ICM thermal emission at lower energies, and lacking spatial resolution at the upper X-ray range. The distribution of radio synchrotron radiation can provide great insight into the particle acceleration mechanisms operating within a cluster.

Diffuse radio synchrotron radiation has long been observed at the cluster scale, up to and beyond 1 Mpc at the largest extents. At these scales, radio emission is generally categorised into radio relics and giant radio halos. Radio relics trace shocks, while radio halos are likely to trace the merger-induced turbulence within the cluster (e.g. Brunetti & Jones 2014). Given the prerequisite for cluster-scale disturbances, these phenomena are generally limited to disturbed clusters undergoing active mergers, or having recently merged.

However, diffuse radio emission has been detected at smaller scales in typically relaxed cool-core clusters (e.g Giacintucci et al. 2014). This "minihalo" emission is typically localised within the cool core, with typical radii from 50 kpc to $0.2R_{200}$ (a standard metric for determining the boundary between the core and the outer ICM). Giacintucci et al. (2017) found that these structures are not uncommon in the high mass subset of relaxed cluster, with 80% containing radio minihalos.

— Relaxed clusters can have gas sloshing in the core causing turbulence in the ICm. An increasing number of these relaxed clusters have had radio minihalos detected within them.

Radio minihalos are diffuse faint synchrotron sources, stretching across $0.2R_{500}$ to hundreds of kpc. The size of these sources is well beyond the scale expected from the lifetime of particles created from a singular source, with these CRe instead created in situ. There are two main theories behind the local production of the highly relativistic electrons, hadronic and turbulent re-acceleration. Hadronic production occurs due to interactions between thermal cluster protons (from the CMB) and cosmic-ray protons, causing a cascade of byproducts including a highly relativistic electon. In contrast, the reacceleration method supposes that stochastic acceleration is applied to a lower energy population of CRe, reaccelerating them to the ultra-relativistic levels found in minihalos. These secondary particles nmay be sourced from AGN, hadronic, etc and have a longer lifetime and therefore length-scale.

There is some evidence towards bounding of the minihalo emission with the cool core of a galaxy cluster. This boundary is often characterised by a 'cold front', where the denser cool gas of the core interacts with the hotter diffuse outer ICM. This

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implies a connection between the properties of the core plasma and the minihalo emission. Recent works have blurred this understanding, with much larger extents outside the cool core observed. This could be evidence towards a multi-component system with different mechanisms at play across the scales of the minihalo.

RXJ1720 is a neat cluster. it's got a whole bunch of papers and is one of the originally detected clusters. The majority of the observational work has occured ¡ 8 GHz, though the robustness of the 8 GHz measurement has been called into question. These studies found that the minihalo exhibits a central component and a tail, with spectral steepening along the length. The 8 GHz measurement appears to show spectral steepening at high-frequency.

Here we present new 8–12 GHz data, highest frequency observational data for this cluster. Few similar examples at this frequency. This paper is part of a 10 GHz survey, to explore the usage of this data for more deeply understanding the likelihood of each production mechanism. We also take a look at some archival L-band data at 1.5 GHz, since it doesn't look like it has been published yet.

We like LambdaCDM with rounded parameters because its close enough. These are the parameters for 1" = a few kpc. We also like J2000 coordinates, since they are closer than B1959 or whatever.

- Introduction to the galaxy cluster RXJ1720+2638 and its general properties, ala Yvette's paper
- Introduction to the minihalo, what is a minihalo.
- What are the proposed methods to cause said emission, with references to the simulations.
- What is this paper focussed on: 8-12 GHz observation of the minihalo. What parts of the paper correspond to what thing.
- Mention G14 so that we can mention non-weirdly? during analysis.
- Coordinates + cosmological settings assumed etc

1. Observations

We present new VLA images at 10 GHz, allowing us to image the minihalo with greater sensitivity than the 8 GHz observation analysed by

- The primary 8-12 GHz dataset info (VLA, configuration, dates, observation proposals, etc)
- What primary and secondary calibrators were used.
- The L-band observation

2. Data reduction

- Description of the data reduction steps.
- Using CASA pipeline to calibrate the data, with a modified flagging step using AOFLAGGER instead of the CASA version.
- Mention broadly the steps taken in the pipeline.
- Reasoning for above is better quality flagging for the microwave emission in bands, due to Elon Musk or something.
- Note where interference takes place, alongside how much data was removed from each observation.
- Imaging done via tclean with broad nterms / weighting parameter. Leave the subtraction and stuff for analysis.
- Separate items for L-band and X-band

3. Analysis

3.1. Subtraction of AGN

- Checking the difference in measured flux for the point sources since measurements C & D are a year apart.
- Some statistics on variablity
- Uvrange for cutting out diffuse emission + point source modelling.
- Show results, compare results of secondary calibrator.
- Conclude its probably fine.
- Since variablity exists, each dataset AGN was modelled independently then subtracted.

Image of point sources, labelled AGN, head-tail, right (better name)

3.2. Subtracting the right galaxy thingy

- Difficulty in getting a clean image due to a noisy point source to the right of the minihalo $\sim$ beam size
- Also subtracted this using the same process above.
- Little information about the source, just mention brightness, size, location.

3.3. Calculating minihalo flux

- With clean image can now calculate the minihalo flux.
- In order for comparable results, estimated mask from G14 to use for analysis.
- Computed total flux within the regions, then the error.
- Show derivation of the error, justify differences in error calc to G14.
- Might as well show stability to robust settings [1, 2] if we have space.
- Show possible re-energisation, but below 3x sigma.

Image of minihalo with mask contours overlaid

3.4. Comparison to other frequencies

- Compare the points above to the G14 data collected + things from Y21.
- Show agreement in the full, centre, tail, AGN. Plus comparisons between them.
- Lack of steepening.

Recreation of plot from G14, with additional information

3.5. SZ effects

- Look the contribution is negligible.
- This is how we worked it out: Little thing is littler than the little thing we are measuring.

3.6. L-band data

- Not entirely sure what to do here.
- Also don't have a perfect image yet.
- Point at the possible gentle re-energisation also present here.

Image of L-band minihalo

4. Discussion

- Discuss in detail the difference between the hadronic and re-acceleration scenarios.
- Talk about the conclusions taken from G14, plus the new data collected since then, in particular the presence of sub-kpc jets Y22 and Y21 results.
- More discussion about differences between the tail and centre.
- Actually do some reading.

5. Conclusion

I like the bullet point style of the main points of the paper, so might as well do that.

- Clean imaging of 8-12 GHz VLA observation for RXJ1720+2638 minihalo to measure minihalo flux.
- Calculated flux density of the minihalo, tail, centre.
- More things from the discussion

References

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